

ActInf ModelStream #004.1 ~ "Implementing Active Inference by Message Passing in a Factor Graph"

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a signal processing group and so i designed my training sorry actually i i just one thing changed could you just restart that but go for it okay uh thanks for the introduction then my name is bertie fries i'm a professor in eindhoven at eindhoven university of technology in the netherlands i'll be a part of an electrical engineering department in a signal processing group so we designed signal processing algorithms and about six seven years ago i read for the first time a paper by carl fiston it was called a rough guide to the brain and struck me that this could be fantastic for signal processing so since then i'm really trying to work with people in my lab on realizing realizing agents that will design will automate the design process of signal processing algorithms and as you know we're doing this by message passing and we want to talk about that today uh thais yeah um my name is tyson i'm a poster with bird's lab i did my phd also an active inference on how to automate those processes and uh also together with michael cox and all the colleagues from the lab we built the toolbox called fornlab and i'll be talking and walking you through um how we apply that in an active inference context um and do some cool things with that and so that's for later yeah hello everyone so my name is nitri bhagav i'm a phd candidate in bias lab also in hue in university and yeah my work is mostly about reactive message pricing based bayesian inference that we hope will help with active inference as well but that's also for later i will talk about it on my uh time yeah my slot cool thank you okay um shall i uh go then and do a little introduction a few slides let's see that yeah okay um let's see if i can share my slide yep looks good yeah okay so the first slide is uh eindhoven um because you may wonder where is i know in relation to amsterdam well it's about 100 kilometers south of amsterdam close to the belgian border and not so far away from germany either um so it's a it's sort of a high-tech city phillips originated and opened on the right bottom you see a picture of the center and the right top is a view of the campus of our university of technology here's an aerial view of groups going through five here's an aerial view of our campus and uh so here let's see if i can share a pointer yeah so uh this is the

building for electrical engineering so this is where we are ios lab is short for bayesian intelligent autonomous systems that's what we try to build we have about three let's say staff members faculty members and currently six phd students dmitry is one of those pc students and we have open positions if there are people watching that are interested in probabilistic programming or how to make active inference work then what are we trying to do this is a picture that's probably familiar to everybody in this uh forum right this is uh this wants to show that well the only thing that's really going on in the brain is for energy minimization or expected free energy minimization to do everything and that's a huge inspiration to us to us engineers so what we try to do is basically this we want to put this in an iphone or on a raspberry pi and let a robot learn how to ride a bike but the beauty of this framework for engineering purposes is that it's almost one solution approach to to to any problem so if we can do it you could teach a robot how to write a energy minimization probably we can also apply this in virtual reality and design algorithms for hearing aids or even self-driving cars the the big let's say promise or the attractiveness for engineering is that it it's just one always the same thing you just have to propose a model and minimize the free energy no matter what the application is but it's very uh appealing the problem for engineering is that this energy functional is a function of observations and observations are streaming data coming usually well could be at every millisecond so it's a highly time varying function and the number of latent variables let's say the space of latent variables is usually very high so we have a very high dimensional function that is time varying and we want to minimize that now the brain is very good at it right the brain has uh what is a 10 to the power of 14 synapses 100 billion euros but a normal optimization library and model up or python will not cut it you cannot minimize a time varying function of thousand variables in matlab it's not going to work so we need something quite radical here and the idea we go with is not that radical again we just take inspiration from the brain right the brain is a network of message passing and turns out in my own field in signal processing or information theory to be exact this has been formalized there's a paper here by dave forney in 2001 codes on graphs normal realizations he called it normal factor graphs but in honor of david forney there today they're called forney style factory graphs so this is the origin of the factor graphs that we are talking about and a few years later hans andrea lalger at the university in series uh i have made this popular in the signal processing community and already around 2007 you see in in his papers these uh yeah these typical structures like this that we will show also later these are kalman filters this is really also what tristan talks about these kinds of structures so um this is uh what we want to

do in our language is to combine what active inference from the inspire inspiration from the neuroscience community and combine it with what we know in signal processing and information theory about factor graphs and use these tools to to implement this uh so today we have two presentations one by text from the line uh on how to do message passing by forney lab forney lab is the toolbox that we wrote uh you know the name of course refers to the forney uh and thais will show how to do active influence with this toolbox then we're also working on a new version it's called reactive message passing and dmitry is the main person there and dimitri also i'll talk about that a bit and that's all for me um i'll give it back to you dan or maybe uh can continue what is your preference i'll stop sharing my screen see super interesting thanks a lot for the context and it will be awesome to hear from the authors with maybe any reflections on those general points you raised and how it gets applied in a really specific way because you brought up a ton of topics that we talk about all the time like the simplicity of one unified approach and the challenges of ongoing optimization problems and drawing inspiration from nature so great things i think we can just jump to whichever of the presentation you all prefer all right then i'll go ahead and try to share my screen see if that works [Music] select windows screen entire screen okay a lot more practical streams i think you can see my screen now yep you can make it look just large on your side and i'll resize it there we go looks great there we go all right well thanks again for having us can you click away i will be giving a little introduction sorry where it says you're sharing your um screen you could hide that little jitsi thing um and it's gone thank you so um yeah i'll be giving a little introduction on how to do message passing with phony lab in the context of active inference and to give a little bit of motivation for this talk um this is kind of the situation where we're interested in so we have some kind of an environment and we have an environmental process that's running in that environment and we want to develop an agent that has some purpose or does some purposeful task within that environment and the agent is allowed to send actions to the environment and manipulate that environment and it will receive observations and in the environment process there is some function rt that is running there and it might be simulation that we run that we interact with that or it might be even a real world process of which of course we don't really know what rt is running in the real world and somehow we want to build an agent that does something in that world so where do you start with that there's this paper from conant and ashby in the 1970s and they have an interesting theorem that says that every good regulator of a system must be a model of that system that actually means that if you want to build an agent that regulates or manipulates our system the environment then we

have to model that system so as an engineer we are in the business of building models generative models to be precise and this model represents our belief how observations follow from our manipulations we represented by a function f that is a function of y the observations the controls u and also some latent state hidden variables x that act as intermediate between observations and controls now reasoning forward is one thing but in the end what we want to do is observe things and then propose controls that lead us to favorable states and that is where the idea of the free energy principle and active inference comes in so if we want to see this is the hypothesis so that if we want to build an agent that does something purposeful then we can do that by minimizing a free energy bound on surprise so we want to build an agent that avoids surprises and by that we can do bayesian inferences about that environment so we have to build some kind of free for the agent to optimize and by that duration inferences for actions and controls and we defined as for energy functional as a kl divergence between the generative model that we have and some kind of an approximate posterior that we postulate that we here define q t and minimizing this energy functional will then allow us to reason backwards from observations towards controls that's kind of the general idea having postulated this i have choices to make so for example one big choice that i want to make is how do i choose my model how do i choose my f in such a way that is useful to me and also how do i choose my q t but how do i choose the factorization of that such that i can make my inferences that process is kind of a thing of trial and error and um as an engineer i want of course to build the best model that i possibly can but how how do i go as quickly as possible through a process of trial and error that gives me the best mark the best model and this is where the idea of a model design cycle comes in and this was again made popular by david bly in his paper in 2014 built compute critique repeat and he proposes a cycle where an engineer proposes a model and then with that model you want to infer your quantities of interest given data and if you have infer the model like that you want to criticize it and based upon well the performance that you have evaluated you want to reiterate rebuild your model infer again and see how well it then performs until you're satisfied and you can apply it in your agent in in practice maybe in a real world setting and then the challenge becomes then how do i go through this design cycle as quickly as possible so we want to be flexible we want to automate things and making model proposals itself is something that well you have to do as an as an engineer you have to come up with a proposal of how you believe the world functions but once you have that everything else can be automated you can infer these quantities by probabilistic programming you can evaluate the free energy as a measure of model performance automating this cycle will be the

key to making model proposals for an agent that will be useful in practice so you want to go through this as quickly as possible so that's why we choose a specific method we choose vector graphs to represent our factorized models because in a graph you can do manipulations you can add nodes delete nodes you can rewire things very quickly and once you have a graphical representation of your model you can do message passing on that graph in order to automate the inference and you can even evaluate free energy by local contributions in that graph so that is why we choose a graphical representation and that is why we want to do message passing because we want to design models quickly so that we can design effective agents now how do these graphs kind of work right how do you build an effective graph or what does that look like so we choose a specific representation of our model and this is an example where we have a generative model of five variables x_1 through x_5 and we have three factors f_a , f_b and f_c and here on the picture on the right is the graphical representation of that model and as you can see the edges correspond to the variables and the nodes correspond to the vectors and the edges are connected to the nodes of which the vectors are in arguments so here you see node f_a which connects edges x_1 and x_2 well that is because x_1 and x_2 are arguments of f_a same for f_b and f_c how do we do inference in this graph so suppose that we observe x_5 which is indicated by this little solid square and we want to compute our belief over x_2 given x_5 well what do you do well you marginalize everything and by marginalization i mean you integrate out every variable except the one that you're interested in so accept x_2 you integrate everything out and you add this constraint that you observe x_5 now if you have a big model then this integration becomes very cumbersome because this here is you have here four variables of which to integrate this room this integration space that really explodes if you want to do this naively but you can be smart about this you can re-shuffle these integrals according to their respective terms so for example this integration over x_5 well there is only one factor where x_5 is an argument of so i can use the distributive rule to bring this integral inside so that i divide these integrals up into smaller parts which are manageable and this is where message passing comes from message passing is essentially solving these integrations one by one so i can first integrate out x_1 here for example which as we say summarizes all the information inside this orange box and it gives us a message first message here that exits this orange box then it can continue for the bottom so integrate out x_5 and i get a second message and i can use this second message in the computation of the third message so this way you get a nested solution approach where i use the the the solution that i have for the message two in the computation of message three and you see that in the

end the multiplication of these two colliding messages on the edge for x_2 will give me my proportional belief my proportional posterior over x_2 given what i have observed so to give a little example of a specific node how that works here's an equality constraint node which we use as as a constraint to to constrain the beliefs upon the the adjacent edges to be equal um and what that does is that they will follow if we if we go through the math here this is what the node function looks like so i say okay it's a function of three variables x y and z and i constrain z to be x and i constrain z also to be y then x will also be y if i constrain those two to be the same so let's see if we can derive messages based on this if i want to summarize the information into this orange box i can use the sub-product rule for that i have the two messages coming in here from the left and from the bottom message one and message 2. i multiplied that by the node function of the equality node and i want to perform this integration now i substitute this this node function here with dividing integration and then i can use the shifting property kind of replace the arguments here of x and y with the arguments of z so what i have essentially done here is saying i can compute this message three by multiplying the two messages that are incoming from the left and from the bottom now if you squint your eyes a bit and you look uh you look through your eyes you can kind of recognize a proportional base rule in here so essentially you can say well if message three here represents my then message one can represent my prior a message to my likelihood and then i have a proportional posterior message three and then i can pass on to the rest of my model and this node is also often used to kind of combine information that's coming so this is a very quick introduction to do message passing um we have derived lots of rules also from the literature also implemented a lot of rules that are already derived into uh into phony lab which is basically well a software problem programming suite that does the scheduling and this message computation for you so you don't have to think about that too much how to redistribute these integrals and how to derive well a lot of updates for specific nodes um so that's kind of the things that that we can we can automate uh on that side of the uh of the design cycle other any questions about this because i'm going through this very quickly and what i'm going to do next is kind of want to walk through an example so maybe this is kind of a natural point where people have questions that they can kind of stop me and think hey you're going to fast here i think it'll be good to see the example and then we have some questions in the live chat and also anyone's welcome to add more all right sure sure so how do you do this in practice in an active inference context so this is a little example it's called the bayesian thermostat where we have an environment environment where we have a heat source here on

the left and we have a little car that can move around it can move away from the heat source or it can move towards the heat source now the position relative to heat source is what we call x and at every position there is a specific temperature and temperature that we measure at a specific position is what we call y and we have a preference of being at a temperature of four y four i don't know it's something we choose now we control control this car by moving it left or right and our control this is the velocity that we move left or right with and it's called u so this is our environment and we want to build a model for that in active inference what we want to do is we want to reason from states where we would like to be towards controls that we have to apply at this current at this current time in order to get there in the future so what we want to do is we want to reason forward from the current time t up into some time horizon t in the future we want to consider how the agent will move and where we want that agent to be at that time and in the end we will want to be at a temperature of four and that's a state space model that represents our belief for how this agent will move itself through the world so at time t we have an observation coming in of a specific temperature and maybe we have already made an action u_t and we say that the current state x_t relates to the previous state and action that we have by adding so basically u_t here is a velocity that we add to our current position x_t in one time step and then we get kind of our next position by just adding that and also applying some gaussian noise and so we have a state transition that is this additive with u_t with duty saying i want to move forward to the left or four to the right u_t and then we add some gaussian noise saying okay we're kind of uncertain about u_t how this agent might move in the environment we have an idea about that but let's add some variants in order to account for some uncertainties and then we say here with this vertical line that if we have u_t an observation or if we have a position then it relates to an observation of the temperature well -1 it's kind of a rough guess it's just saying well if i move to the right if i move away from my heat source then my temperature will decline with a slope of minus one and it's kind of a really rough guess in the real world there will be a very nice temperature gradient that maybe will move like will move very slowly downwards u_t but we're just saying hey we don't really know what that is let's make a rough guess we have a belief this is our generative model our belief about the world let's just say that temperature decreases with position and again we're not very certain about that so let's again add and now i extend this into the future so i say actually well i believe that in the future my environment will also evolve according to this but i also have some ideas about where i want to be and this is where gold priors come in that constrain my future observations of the temperature to be around this desired temperature of

four and i'm saying then here with some added noise well um i want to be around four right it doesn't have to be exactly but i want to be pretty certain that um i will be around four at time t plus 1 and i will continue to remain there until sometime in the future now we have regenerative model we can define our energy functional and we can solve this for energy functional by message passing on this graph and we pass these messages towards our next control because we're interested in inferring what we have to do next like from from from the next time um and this is where we can do message passing we can summarize all the information that we have from from the past so this will be an estimate of the current state of the agent and it will be biased by our beliefs about the future saying where we want to be and that will influence or determine which control that we will take at the present time or at the next time all the details of that this are are described in this paper from 2019 by you can also see kind of how we apply within the action perception loop and things like that because you will have to do this at every time t because your state estimate will change the word will change according to your actions so you will for every time you will have to recompute a new action for what you're going to do next um and this is kind of the the main idea now of course we don't want to do everything of this by hand so what we want to do is well we want to use a tool for that so now i'm going to show a little demo of how you would implement this um using the the formula probabilistic programming tool so let's see if i can open our little demo so this is uh this is readable actually or should i zoom in more you can zoom in a little bit more maybe like this uh yeah i can resize it that that's pretty good oh that's perfect that's pretty good yep thank you so this is um so so it's kind of set what what we want to do but so this is kind of our definition of the environment itself so this is the the real world well we have this this temperature gradient which is a very nice function so very smooth function where at the heat source we have a temperature of 100 and it it goes down with with with position from the heat source so this is a real world we don't really know this so this is the one that we're going to approximate with this very rough minus one and then we'll see whether that will work on it now this is my wonderful ascii art that has now been resized of basically the model that i've just shown in the slides and this here is our model definition so actually i have to zoom out a bit to make it more readable yeah so we want to build a graph for this right so um here we have our observation states controls and we define a prior belief about t minus one so where have we been which we're going to say is a gaussian and with some mean and some variance and then for every time into the future from uh from one until the horizon we will have a prior belief about controls which basically says well what am i allowed to do when i say okay i have

a i'll have a gaussian prior about controls with a mean of zero i'll substitute that there and then maybe some variance here i say well i have my position of the agent it relates to my controls by just this addition with some added precision gamma so this is our transition model this is horizontal line and this is our observation model which is very rough estimate minus one of the temperature gradient times current state and also some some some precision conservation precision and this represents my goal prior so note that i haven't really put in any specific values here yet i've just have placeholders for actual statistics and values that i'm going to put in because the actual derivation of my algorithm doesn't really depend upon the statistics it's just something that we can we can put in later so this is what i want to do build a free energy functional and the phony lab is actually just basically four lines of code um so this stereo factorization that defines my factorization for q saying that um i'm just going to have a joint variational distribution um over the entire graph i don't really subdivide it into a structure into a structured factorization some people are familiar with that um but just saying okay i just have this stereo factorization of the variational distribution to be my entire graph and never just one command i can derive a message passing algorithm that propagates all the messages towards control u_2 which which which is the control for the next time step t plus one and i want to evaluate my free energy so also give me at the same time an algorithm that i can i can execute in order to evaluate my free energy and this line will convert this two source code that i can then load in my environment and what this does is it builds a message passing algorithm for us so this is this this is the code that executes the message passing on our graph and here you can see that it computes all these messages by some product rules and other rules there we have pre-derived and we have implemented in fornilab and you can see that these messages they depend upon the statistics so this is for example the mean of of our prior over states and variance of a prior overstates and we have other messages that depend upon previous messages in the way that i explained message passing and this builds an entire list of in this case 26 messages and in the end we're interested in the marginal beliefs so the posterior marginal beliefs about about for example or controls and you get these beliefs by multiplying messages together so for example to get a new belief over the following control you multiply messages 9 and 25 and you also recognize this from a little introduction past introduction where you have colliding messages on the edge that you multiply in order to get your posterior beliefs and that's what we return in the end that's what we're interested in so this is the execution of that algorithm and notice that we've made an entire schedule here and they can be very cumbersome can be very long very long schedule and so that makes it expensive maybe

to load and to execute and dimitri has the solution for that and he'll explain that in a minute um so i'm very happy about building schedules and um dimitri not so much uh and he tries to find uh find a solution for them get rid of them um because in the end that will be the fastest to really don't have a schedule that not really had recipe but just cook by whatever is given to you so now we execute this algorithm in practice so this is our action perception loop for every time step we want to we want to act we want to evoke an action from the agent we want to execute that that that action in the uh in the environment so this is sending the action towards actual real world or kind of simulated world in this case and what we get from that is an observation and from our action observation we want to infer infer our new action and also return the variation of free energy and then prepare for the next step so in every step we propose an action to the environment it actually gets executed we observe a new outcome we we infer new action um and we evaluate how well we did with that action and then you get well nice plots like this so this is the velocity of our agent we start at zero and you can see that it moves very quickly to the right and then it also moves back a little bit um until the end comes to a stationary point more or less um and you can see actually here what it does so it starts out at a temperature of 20 then it moves very quickly to the right away from the heat source it overshoots its mark and then it actually goes back and then it settles around this desired temperature of four that we have uh encoded by by our goal priors um so why does overshoot well our model of the environment wasn't perfect right we had this very rough estimate of minus minus one um and in the real world was this very nice smooth kind of complicated function that has this kind of bell shape so even though our generative model of the environment wasn't perfect um in the end we paid by that pay for that by overshooting a little bit but still we were able to get to our desired state at the temperature of four so in the end it kind of worked pretty well and you can also see that the free energy of the agent also decreases with time so we start off at around a thousand so the scale's logarithmic here and it decreases very quickly um to our sum value it's rather rather low right and then we have some noise that that induces some unexpected changes or some unexpected surprises um so that's why you have these little ripples these little ripples here but in the end we're minimizing free energy and it's going it's decreasing drastically so we actually have this free energy minimizing agent and yeah that's uh that's basically all i have the story and and the demo that was awesome thank you very much we can uh with screen share on or not ask a few of the questions from the live chat and also give people a few seconds to type more so i'm just going to jump into the questions first let me stop my screenshot great so we're still sharing the screen

story um we see just the jitsies so go just click the screen share button again in jitsi so awesome thank you so the first question is from john and john writes how does the factor graph approach work if the graph structure is not known beforehand is structure learning possible uh yeah so so that's a good that's a good question so is structural learning possible in graphs like these so that is actually something that is still an active field of research that would be awesome if you can automate also the structure learning of the graph itself but there are a lot of challenges there so how do you parameterize this graph structure um or and and how do you do how do you do that actually right what is your search space in terms of graphs how do you how do you learn node functions if you don't have a node function given for your graph right so these are all very difficult questions and you don't have a straightforward answer so so that's why this actual model design step in the design cycle itself is still kind of a creative process there is an engineer that has to come up with a model and as an engineer you also have to think about how to adjust that model if you're not satisfied with it so if free energy is still high after you run your agent well something is going on something is wrong with your model it's not an accurate representation of what is the environment and then you might adjust that how do you adjust that well currently it's still just trial and error so maybe you can say okay i think here is still something that i can improve maybe remove a node maybe the model is too complex but structure running in itself is um it is something that is also it can be addressed so there are there are methods like um like nested model comparison and things like that that you can do in fact graph so you can for example if you have a nested model compute savage dickey ratio and then kind of prune your model based upon that and so those are things that you can do to find a complex model and see what you can cut away but adding to that model is still difficult so once you've cut it away well how do you add yeah i i don't i don't really know so that's that's a good question and uh i don't have a very straightforward and switch you can from my rattling so yeah perhaps i can say something about it um i mean i i totally agree with uh with taste i mean generally when you design a system or signal processing system you have to design the structure uh you have to estimate the parameters and then you have to infer states right the states change very fast uh it leads to a common filter and and that's very well known in factor graphs then the next stage would be can we also over the long term learn parameters and our effectographs can do that pretty now the the next stage in biology i mean we know the structure is also alert over even longer times right and so that's not working at the moment in formula you can compare like tests i says if you have two examples you can compare the free energy and just pick the best one but we are working on that and this is i mean we're taking

again the lead by uh from carl's idea on bayesian model reduction but how to implement that in a factor graph is a research project so we're we're we're working on that and i'm not sure where that will end but that's our goal did you have phd position available bert so also to bring it back to the example of motor car getting to an optimal temperature it's like if every hour you notice that there was a bump in the free energy then you go back to the drawing board and you you go to the people who are on that car in that area and ask well what happens every hour or maybe we do need to include this other source of information but the model that you had does its job so you could include the wallpaper and other features the room but that's kind of this art and science of engineering and that's why it's always interesting to work because often in research it's like let's finish the analysis but really there's this cycle that the entire model is embedded within that helps us always keep an eye out for those patterns that are precluding being captured by our current models here's just here's a second question from the chat how does the factor graph approach work if the goal temperature in this example is not defined beforehand can the goal be abstract so kind of a similar question but rather than structure learning how does preference and goal orientation arise again excellent question that kind of relates to the question well who sets the goals or how do you set goals yeah the plain answer to that is well it's an engineer that defines that um but it can also be a higher level that does inference and sets to his goals so then you get to kind of turtles all the way down argument in the sense well you can put layer upon layer upon layer upon layer and layers set goals and other layers as the goals for that layer etc etc so how that will work in practice um yeah that's still again um active research area but in the end it should come down to minimizing free energy so so that's kind of the central theme of this we want to we want to minimize free energy we do it by perception we do it by model learning um but also learning of the goals should be driven by minimizing free energy so you should choose the goals that minimize for energy well how do you do that i don't know you tell me right one more question that was asked was how or does this integrate with graph databases or with large data sets is this something that scales or is already integrated to work with that kind of empirical data or what um i think it depends on what you mean by large data sets when i hear large data sets i usually think about big data things like that so um this toolbox is specifically built for dynamic modeling right so now in the sense that you have a lot of features in your data but it's signal processing so data might come in very quickly and you have a model of how these data change over time in that sense you want to do processing quickly so you want a lot you want velocity it's it's based on processing with velocity not processing with with volume as for

example other problems programming toolboxes are excellent at like um by torch and things like that right sampling base toolboxes adword all great toolboxes if you have a volume in terms of data and you have for example an iid model that you want to uh that you want to fit um in our sense if we talk about large data we talk about volume so data that come in very quickly and we try to be good at that um phony lab is not yet that good at it to be honest um but reactive message passing is going to be the bomb at those kind of of data sets there's there's also a bit of that what we we really try to build these toolbox to build active influence agents and the idea that you just put these agents in an environment they make actions they select their own data so we don't really try to build a toolbox that is really good at just doing machine learning for a fixed set database but rather we try to build models that let's say dynamic models that can adapt and that can make actions and that can in real time process streaming data and and in that sense over time if you wait long enough yeah there's a lot of data streaming through it but it was sort of generated by you or at least affected by the agent itself right we don't try to optimize for for big fixed databases rather for just streaming data in an environment it's a really fascinating point because there's a lot of time dependent tasks like autonomous driving where it's like okay here's the 500 terabytes of video or here's this now give me the best possible score in a time snapshot and it's it's every single time with an active inference agent we're always including action and policy selection in the loop and so how how are we going to deal with that but that's why the inspiration from biological systems becomes so important it's kind of like when people go oh well the brain must have this processing power and then if it were a light bulb or if it was this kind of a silicone computer it would take up this much energy ergo's efficiency is such and such but it's predicated upon the kind of computers that we're used to seeing instead of kind of starting with what already exists and then looking for an answer that rests in that rather than how do we make lower power transistors so that we can fit them into the brain awesome topics i think we're ready for the next presentation yeah hello everyone let me also share my screen uh for a second i hope it should work now looks good so can you see my slides now yes so okay so yeah hello again my name is mitri bhagav and i want to introduce you like this new reactive message passing based framework for bayesian imprints in july that we called reactive mp it's more like our future vision on solutions for active inference fornilab.jl is like more like mature framework it works and reactive mp is our research project for now and we are going to start with a what is actually reactive message busting and in a in most senses it's the same message passing as a formula but re-implemented in reactive paradigm so basically the major issue with traditional

approaches in porn lab is that enable in order to run inference for our model we need to create this shadow of messages in advance so we need to pre-analyze our graph we need to analyze everything basically and we do we need to do it every time we change our graph structure and if graph is big this graph analysis time and shadow creation may take a lot of extra time and it might be not a big issue but as an engineer you may want to test a lot of models until you're satisfied with the performance and this extra delays and shuttle creation times that they may be just a bit of annoying and reactive message passing last allows us to eliminate this predefined shadow and also it gives us a lot of other benefits and possible future research directions so instead of having instead of having a fixed shadow we cast our graph to like an event-based system where everything can react on its neighbor environment and in terms of message passing nodes react on incoming messages and also it reacts on updated posterior marginals over our parameters and the whole model becomes reactive we can also react on changes in this system and basically do whatever we want so we may perform some action based on new updated posterior marginals it also in message passing a natural starting point uh is uh our data or observations and the model itself reacts on changes in our data or in our priors and it also changes posteriors accordingly so here i outlined some extra benefits from the active message passive implementation so first of all we may outline biological plausibility because in nature we probably don't have any predefined scheduler for our information flowing on neurons in our brain for example uh they they are driven like by some chemistry like physics and in sense it's like also reactive system it constantly adapts the changes in an environment and also it reacts only if needed and in terms of message passing based inference it may be unnecessary to even react in some events like some messages if they aren't really important uh a fixed predefined shadow you're forced to do all computations even if you don't really need them uh another great benefit of craft message busting is that it scales very nicely for very big graph so it's more like an answer for previous question uh so it supports like hundreds of thousands of factor nodes even millions are possible uh i will show you an example in a couple of minutes but just to give you a bit of context so in this plot we met we modeled just some moving objects with known line linear dynamics and the idea was to estimate the sedan states given noisy observations in this example i used kalman smoother with 50 000 observations and this model contains roughly about 150 000 factor nodes so bayesian inference on this kind of a model is impossible on sampling based inference but with reactive message passing it takes only about like eight seconds on just a home macbook laptop so we can go further a bit and make reactive system robust and tolerant to some let's say failing nodes or missing from let's

say failed sensor we fix predefined shadow if something like this happens we need to do everything from scratch we need to recreate our model we need to again make a shadow and it will just take some time with reactive message busting we may just stop reacting on some missing data or maybe failing sensor and we can just wait for them to be available again it makes it very robust in that sense it also gives us an opportunity to change our graph structure in runtime and still perform inference without stopping so there's a slide about how we actually we use julia's main programming language in our lab roughly about a year and a half i believe ago we wrote the library for reactive programming in julia so it's completely unrelated to message passing in just a general framework for event based systems but that allowed us at the later point to build reactive mp.jl package and it implements free energy minimization by message passing we also introduced graphpl.gel package that is a high level and user friendly probabilistic model specification language that we use in our demos um yeah here i'm i want to show you an example just in a second so you can you can see my screen right yep maybe one little size larger or two i hope it will not break yeah make it a little larger and then if it looks unexpected you can pull back okay here we go better yep thank you so here is our example so assume we have moving object uh it has some hidden states and just assume that we know it's linear dynamics for simplicity and we don't we don't have a direct access to its location but we have a noisy observation of this moving object and we wish to estimate the true location of this moving object by only observing its noisy measurements and we may use a linear multivariate gaussian state space model like equations looks like this these are equivalent notations they're the same and basically here we say that uh okay we have a state x at time step k and it depends only on the previous time step through some linear operator a and we also have a transition noise which is a gaussian with covariance matrix b and our observations are basically modeled also as a gaussian with covariance matrix q so basically this is our model and we can simply create a factor graph out of it and as you can see our model specification resembles very closely to the equations defined above so here we have our state x_k and it's modeled as like a gaussian over previous state with some let's say known covariance matrices and this is how we build our model and under the hood this code it generates factor graph uh and we can we can all later on use reactive mp.jail api to estimate hidden states of our system but here's our example it looks a bit off probably because i zoomed in a bit but it's fine i believe and in this example i perform a kalman smoother for 500 points and i just decomposed the trajectory of this moving object into two axis x-axis and y-axis and we can see that common smoother predicted like the real hidden states of our system correctly

even though we have a lot of noise in our system so these blue dots it's like our noisy observations and we go we can go further with this so because our system is reactive we can we can estimate uh our our states in real time so here's our example let me try to run it uh i hope it looks smoothly on my computer it's actually very smooth like but i i know that maybe recording is not that smooth like screen sharing but here in this example what we see is that okay we have an infinite data stream of a moving object the blue is a real one and the red is an estimated one and we can see that uh yeah let me reload it because it's kind of off reactive mp is able to perform and bayesian inference in real time and it actually adapts to changes in the environment and also changes the posterior over the current estimated state and we can go even further you probably notice this orange line here just inactive for now but we can also incorporate a prediction for this model so we can just we can just extend a little bit our graph and we can also predict future state of our system like this so now in this example i do bayesian inference in real time and i also predict the future state of my system like in orange and you can see that prediction also adapts to the new observations and yeah it changes basically its future beliefs over future states uh yeah let me stop it and continue with the presentation yeah i'm going to talk about like when reactive message passing is available actually it's already working uh we have a fully working stable backend and api for exact and variational bayesian inference we also support expectation propagation but that api is not yet stable so it may change we support for now only conjugate models from exponential family we support extra constraints for variation optimization procedure such that form or factorization constraint so rocket.jl library it naturally supports infinite data streams for example from an internet or from some sensor and framework itself is able to handle missing data but that api is also not yet stable some of our future plans for this platform so we want to extend it to support non-conjugate models as well basically foreign lab already supports we just need to carefully port all of the available existing functionality from foreign lab to reactive mp and we also want to integrate with other probabilistic programming libraries that exists in july community reactive message passing gives us an opportunity to try to integrate parallel inference and react in different parts of the graph simultaneously using like multi-core capabilities of our cpus we want to extend graph ppl to support modular model specification for now it's unfortunately not possible also reactive message passing naturally gives us an opportunity to run inference where different data streams have different update rates um what that what i was talking about in the very beginning about robustness but it's also interesting how far can we push it and maybe in it's possible to automatically adapt model's graph

structure in runtime based on free but it's like a research project we don't really know if it's possible or not for now uh we we have some ideas about interactive visualization for message passing based algorithms and free energy minimization visualization and eventually we we are planning to release this new platform as a stable version of formula possibly in 2.0 and that basically is everything i wanted to show you in my slides and thank you for your attention i would have be happy to answer your questions awesome first question was is there a paper on this reactivemp.jl or how does that work there is but it's in progress so it's not publicly available but we are working on that so there will be eventually paper about all of this approach i will describe everything how it works and how we use it so yeah i mean of course if anybody writes us an email who wants an early copy of the paper i'll be happy to share it if it stays confidential i'd be happy to share it it's very interesting development maybe to the other authors other um epochs of forney lab were there how did we get with to this reactive programming paradigm and also it seems like you did a lot of the foundational work in the lab on the reactive programming was that just in the julia implementation or in the more conceptual grounding yeah we did actually really a lot of work to support reactive programming in julia basically julia is is like a young language and it has no like good capabilities to run reactive based like systems so basically we build it from scratch yeah but we we used a lot of uh like ideas from other programming languages that we had experience with i mean i know that the active influence community uh matlab and matlab is fantastic but for for really for if you want real time processing of streaming data then julia is a very good is a better option because julia is almost has the syntax of matlab from a user's viewpoint but out of the box it's almost as fast as c and so it was a better combination for if you are in engineering and you actually want to build systems that run in real time you want to make demos and julia is a better language also uh dimitri also makes use of some really advanced stuff in julia it's called multiple dispatch but it doesn't matter what it is but it's it's quite advanced and it's not uh available in in matlab but it's extremely useful for what we do for do with the toolbox it's it seems to be if you look at a graph and message passing it see it seems to be like oh this is actually not so hard to implement but there's a reason why there are extremely few toolboxes for factor graphs okay microsoft has built one infer.net and it's it's great but it's not really a real-time toolbox um and so as far as we know we are there's not much competition at the moment for us not that our toolbox is very advanced yet but um i don't see a lot of people working on this i see a lot of people working on multicarter sampling but multi-car sampling will not work for the sizes of systems and for the real-time data streams

that we want to do with active inference right we are working with systems where we want to make actions and influence our data so this is a real time system and so we have to go to want to actually scale up we have to go build a really high quality professional toolbox that automates message passing i hope this will be uh one of those toolboxes and hopefully will also be other toolboxes but yeah that's why we're doing this very interesting answer how has the dialogue between the math and programming and computer science side and then the active inference side been like what has each side kind of contributed because it sounds like some of those real time insights of active inference are kind of propagating back to the design of algorithms and then we see both directions so how has that played out maybe for any of each or each of you i can also answer maybe this question from my point of view i mostly like doing programming uh but i still learn something new every time because we like in this constant uh dialogue with math guys in our lab let's say and i learn something new every day and of course it reflects like our design choices for our software as well so it's kind of but sometimes it's it's very hard to fit this high level mathematical ideas to the actual implementation and also it's very hard to make it efficient yeah we are not uh we're not done with this um i mean the toolbox is available for anybody and and all the nice territory examples the basic thermostat the tic tac toe and it all works very smooth our hope is that with reactive message passing we can actually you know get a path to scaling this up to to serious applications and that i mean that's real i mean we're an engineering group we want to build systems that really do something useful i also work for a hearing aid company so i want to build real time audio processing algorithms and others may want to use this for robotics so yeah but and that's not going to work in in real time model up it's not going to work um yeah a bit with all the multi-card or something so this is our effort and it's quite it's takes a long time because it's very multidisciplinary efforts right we have in in my group we have a mathematicians neuroscientists and computer scientists because it's it's it's it's difficult there's a lot of different expertises that that you need to build a good toolbox for active inference i think because it's so difficult there is a lot of like um like cross fertilization between the active inference community and the engineering that we do the community is mostly interested in explaining biological systems right the the energy principle and active inference as a model for biology and i think that's very interesting because while nature has a certain way of doing things efficiently um and if this is a model for that then it's also a good idea to take that to engineering and that's kind of our recommend so we kind of take the ideas that are available in the active inference community and where they explain brains and think about brains and

think hey how can we take these ideas use them to build an engineering system and i think that's kind of the main yeah the main interaction between it and hopefully then we build tools that the community can also use then in their research eventually what that made me think about was it's really a reframing of some of the main challenges like seeing signal processing as a real-time event or seeing the causal relationships between action and future data all these kinds of reframings of the problem it's not like the active inference algorithms take massive matrix uh calculations to do per se they can be very simple or not but it's actually that reframing in the the embedding of action within every that then ends up solving the scaling some of the scaling challenges some of the resilience challenges because like resilience of a transportation network in a city they would do sampling or they would take something that's an unfolding dynamic process and then try to roll out a million iterations of it statically so it's again it's not that the calculations inside of the loop have to be challenging it's just a reframing of prioritizing action ended up going down these roads of scaling and real-time capacity yeah i i like that the thought um perhaps i can say something about it in the context of what i work on because i work on designing basically i'm a hearing aid engineer i work on and if you ask a hearing aid engineer signal processing engineer so what is your task well our task is to build the best hearing aid algorithm then what happens is that a hearing aid client goes to the store and buys a hearing aid usually it's very happy and goes out and then two weeks later sits in the restaurant i cannot understand her conversation partner because there's noise and this wasn't expected and there's nothing she can do because you can't ask a client a hearing aid client to fiddle with hearing aid parameters so if this happens a few times um then she throws or she puts her hearing aid in the drawer and this uh indeed about 10 20 percent of hearing aids end up in the drawer they're very expensive and ends up not being used a really sad statistic so rather you can you can turn it around and say okay what is the real hearing aid design problem the real hearing aid design problem is send somebody out with any hearing aid but what do you do when she's unhappy in the field at in the restaurant and what we want to build is an agent that will then so she just slaps her wrist raceway i'm happy and now this agent needs to make an action and give her new parameter settings that are the most interesting for her and and and the in the best compromise between information seeking and goal-driven gold riven being making her happy hearing a design from my viewpoint now is just build an agent that will make actions make hearing aid proposals when she's unhappy and if she right she's unhappy and then hearing a mixed proposal says no that's not good well the agent gives another proposal and that is okay that's better and then we move on and maybe a week later

the same event and this goes on continually but there is at least a procedure to keep moving on to keep improving over time right so that's the the real thing of design is the is then the the action in the field in c2 which is very different from what's currently happening which is hearing aid engineers sitting at the desk with model up not in the environment so that's a it's it's a paradigm shift and active inference agents can maybe make that happen and could really i think uh create a pivoting point for something like signal processing design which is uh you know it's not something that you think about when you're a neuroscientist but this could really mean something big for engineering and not just for signal processing but also in different engineering disciplines and so that's why i think over time you'll see more and more people coming from different fields getting an interest in this uh active inference and the free energy principle uh even even uh non the not neuroscientists right but engineers yeah yeah cool we hope so and and that really reminds me of the pragmatic turn which is it's a little bit like a horseshoe theory with the engineers and the philosophers often talking about pragmatism in different ways different communities different tools and now there's a way to kind of close that gap or at least map across it where we can embed some of those insights about really an activism the role of design science anticipatory design science with take it or leave it philosophy or there's the toolkit to develop maybe it's interesting five years after working with a toolkit or maybe five years of philosophy and then you're curious about the toolkit so one question was how do these graphs account for temporally deep models how are those specified or what make what is different about the graph because it's just calculating the next action to take in the examples that you provided maybe guys you can show your graph again how how the future is represented um and how do you add temporal thickness to that for example that's kind of so if you have if you have a graph chair screen yes if you have a graph like this then how do you add temporal thickness for example um that's that's just about adding layers what is shown here is one layer and this layer it acts at a certain clock time you know observations come in discretely but you might have a layer above this that is somehow connected to this layer below that acts at a higher temporal time scale so it evolves slower but it regulates or influences the parameters of for example a transition model here or an observation model here so you get a time varying observation model that is influenced a slower evolving layer on top of that you can get an even slower evolving layer right and that's that's the way you can build hierarchies and actually there's um also a pg student in our group ismael senos who has done some very interesting work on that and he has investigated how you make hierarchical gaussian filters um so hierarchical gaussian systems where you have one top layer influencing

the variance of the transition model of the layer you can see that you can get can model very natural signals with that so so in nature you have you have signals that are time varying the statistics there their time varying and the statistics of those models are time varying so a hierarchical model in that sense can be very useful for that and of course that's that's yeah of course also how our brains are structured so again there you have this inspiration from nature um on how to structure the models that we try to build with vector graphs at least in theory they should be pretty easy because you can just connect them you can connect layer on top of that and and see how it behaves and eventually the complexity of your model will become very high and you will get penalized for that automatically so there will be a cutoff point um given your data and that will be the optimal amount of layers that you need in order to explain away all the variants that you observe so so that's kind of how we think about temporal thickness and and time dependence and dynamical models like these then can i share my screen also yep sure okay yeah let's see because this um awesome this is a graph from a paper a few years back uh in frontiers um it's called i think uh deep temporal models inspect factor graphs or something and here you see a three layer system i mean don't bother about the details now but here you see at the top layer just just basically has one one section one time step and in that time step at the top layer you have two observations or two steps at the middleware i need uh yeah i'm not sure if you can see my uh okay and then again for the let's say the third layer whenever the middle layer tips takes one step we have two steps here so so there is sort of a finer granularity at each layer right and so we can build hierarchical uh models really and uh so we're doing that also in uh i mean also yeah if you you you look at some of the papers on our website you'll find a lot of babies on the hierarchical gaussian filter from chris matisse we've implemented that in in but also um i mean you can also do it for active influence things like uh and it will look like this uh thomas pyre has also graphs like this in some of his papers very interesting and we see a lot of the same variables like g d b like we can it's kind of a different representation of active inference so this is kind of also going back i think to the earlier frontiers paper what are some of the big equivalencies or kind of like airtight mappings from into forney factor graphs because there's probably some broad areas of application that could almost be hot swapped for this underlying it perhaps so what like are the equivalencies with just bayesian graphs in general versus factor graphs versus other topics at that level i think a foreign style vector graph or a bipartite graph or a beijing network they aren't really different they can be used as they're just a representation of a factorized model and the way you represent that can have an impact about how you think about those models so

for example if you choose a 4-install effective graph representation it's very suited for signal processing because you kind of can see these messages as signals that flow over your graph so for us as engineers this is a very intuitive representation in bayesian networks it's a bit more the model representation itself is more compact in that sense that you only have variable nodes and you see how they relate so it gives you a good idea of model structure so it can be very nice to have a quick overview of model like that and then you have for example a bipartite graph which also shows a relation between variables as an additional kind of vector node in between um and it goes it gives you a bit more granularity on how these are connected and gives you room to talk about well what is the relationship between these variables and so in the end they're equivalent you can take one model and represent it in three different ways maybe even more but i think it does impact the way how you think or how you think about these models so that usually comes up when writing a paper and then you have to think okay what is actually the best representation for my idea sometimes it's about budget graph sometimes it's a visual network sometimes it's for install photograph it just depends by your story yeah hopefully i do find though i mean in my experience when we think about systems and a whole set of equations if you read that in a or if you write it down um it gives more insight if you draw to me it gives more insight if i draw the graph there is an x there must be an exact correspondence between the the graph and the equations and very often if you draw the graph and then you write down the equations there's not an x-axis corresponding so you learn from it you you basically you know there must be an exact correspondence and basically you you often find out you have an error in your equations or in your graph but i think uh like the the end goal the end goal uh is of to make the toolbox a toolbox for let's say for the community to design their own active influence agents and what i envisioned there is something like simulink i'm not sure or labview i'm not sure if people are familiar with that but um these are graphical models you want to actually also define them graphically and you want to have a palette with nodes and just draw your graph and say this is my model and now go run you don't want to worry about the inference the inference is under the hood dependence passing should be done the designers of the toolbox but you should just be thinking about your graphical structure that's the uh and this one and connect it with your mouse to your microphone of your computer or to your camera and maybe to uh you know there's also connectors in the toolbox for robots and then it should just go right that's going to be the bob ross of that's how you want to design i think right you just want to draw the brain and at least the generative model of the brain and just let it go and don't worry about how inference takes

place what what you just said there about viewing the equation and then maybe going from the pen and paper to the programming language to the graphical representation and then kind of cross-checking it's like when you reverse translate languages and you're when you lock in where two words map back to each other you know you've you've made a map but if you're on this infinite loop you're you're lost in the word space so when you move across sectors like that or across modalities with the analytical the simulations on the computer and then the graphical it kind of embeds action in what otherwise might be seen as a knowledge product a product of inference and then a final action and that's a similar fallacy to train the dynamic process in a snapshot and then expect that it's going to work in real time so it's like embedding this real time flow in that the production of knowledge yeah yes yes we are important yeah yeah um yeah we're implementing active insurance while we are trying to design this stuff that's true yeah yeah one one general question was just about factorization the factorization the way we go about it is starting with our intuition and looking for residuals or how of all the ways to factorize a model do we find one that works yeah there's two kinds of factorizations there's a factorization in let's say in the generative model right that's the graph that we draw and so i mean let's say most people uh or the most common models are these these these markov models right where you you kind of retain a current state that summarizes everything that happens in the past uh and you use that current state to basically to yeah to summarize the past and then that's all the information have about the past so you don't need to remember the past you just remember your and with that and you did make a new observation you combine the information make a new state and so forth there so hidden markov palmdps all these models have that same structure these these this microstructure so that's for the generative model of these dynamic then there is a second question if we now do inference um and um yeah then there's often what we call the mean field assumption for the variational posterior but there are also variants on that structured green field so you can you can still decide if you want your posterior to be even more structured or let's say even more factorized than the generative model but in the end yeah it's a it's a proposal you just run it and if you have another proposal you just run it and the one that has the lowest free energy wins that's uh that's it right the challenge is to automate that process go from the let's say the poorer structure to the better structures just by failing to minimization just by message passing in real time without stopping the whole process doing an analysis right it should just keep moving right it should just just structure adaptation should be like state estimation it just keeps moving over time there's no resetting maybe there's a bit of a thing like a dreaming stage right but the principle time moves

on and that's the challenge and we don't know that in our factory class but yeah we i mean we have one hd student who is currently looking into that yeah like as an engineer i think um the question is well what what do you start out with what will be your initial model proposal uh so if you talk about factorizations of your generative model you think uh every factor then it kind of represents a prior distribution or conditional distribution and that's how you kind of build up your model where do you start and it starts with how you believe the world generates observations in that sense you think about what is the causal structure of the environment and you might have some idea about how physics works um or how states transition in in your environment or at least how you believe this transition and that's kind of where you start and you think okay well maybe let's try a very coarse note this minus one for example in my talk let's try that here because i don't really know what i want to put there but i have to put something let's try something and see how that works it kind of models how i believe the world works and that's that's how you start out and then you start thinking well i actually know a bit more about it i'm not satisfied with how this model proposed how this model works um and i can change this minus one for example with uh with something that's a bit more complex because i know the physics i know how these uh how temperature integrates with distance for example and i can build that in um so that's how you go to a second proposal and it's always inspired by uh how you believe the causal structure of the world is um is uh yeah is how that works so in that sense it's kind of theory building you're building a theory or an explanation of your well maybe if you're really really good if you have an excellent model then you can find something that kind of improves upon the state of the art because you can always improve just find something that gives you a lower free energy or better model performance in that sense um and you can keep tweaking and tweaking these models and these factorizations things like that minimization has been described or just as a way to rank the different models and as an imperative on these multiple different fronts how do we know that free energy minimization is making the policy is going to be like resilient is there a way where similar to a local optimization getting trapped somewhere in a bigger optimization space there might be some avenue of free energy minimization that makes the system just like break down like we'll get there faster let's just accelerate and then there's some kind of one-time failure of the system i'm just wondering how can one metric that's a number you know and can be sorted how can it evaluate such radically different chess strategies or driving approaches at the end it's just probability theory and probability is also just a number right and the energy kind of it approximates your evidence which is the probability according to your model of observing the

data so if you make a model that gives you the best evidence then it's a good explanation for your data so in the end it's just you try to do approximate probability theory and free energy is then bound upon your evidence and it also takes into account this posterior divergence term which kind of says the price that you pay for approximating bayesian inference so it has two parts to it it has um the model evidence part and it has the posterior divergence part and both play a role one says okay um this is the quality of my solution it's it's how your model explains the data that you observe you can get you can get a number for that the surprise or the negative log modal evidence and you have the other part of stereo divergence that says well this is the price and information that you pay for making this approximation you can also put a number on that it's just a chaotic versions that you evaluate um and then you add those two and you get a number so yeah so the question is how can one number kind of represent everything in terms of quality well you might care about different things than how well your model predicts or evaluates the observations right and then if you care about other things then then yep then then you might use different number but um in our in our sense in our in our uh in our work yeah we're kind of we want to have this bayesian measure of quality i don't know if that really answers your question but it's it's it's still a good question it's um it's kind of is is free energy or that number that you get is that enough that's that's kind of what's behind this and probability theory says yes but if you apply it in practice then you care about maybe other things you care about how many people survive before how many you know how many how many mistakes you uh you you cash from from something and if you care about that well then you should use that as a performance metric i i think right i mean it's uh the energy of course has has has many decompositions right in in in complexity minus accuracy and and and surprise minus uh minus the kl divergence um but i'm not sure if i interpret the question correctly but if you say like how do you know that it does well in in practice in the field well you don't right um and look the only uh thing you can say is that i mean what the system does it will look for a configuration that minimizes free energy but you don't know what you don't know you don't know if there is another model that would do better if we haven't simulated that model then we don't know the only learning opportunities there are is to generate errors so um the only way to really improve the model is to actually hit indeed um let's say situations where it's not working and then you need to adapt your parameters and and even more over time you have to structure so yeah it's uh i mean building a good system is a process right it's not a matter of uh just you have to be extremely lucky if you build a system that and it completely works right it's what this this between this

orbitals active influence describes is a process towards a better systems but not it doesn't describe a system that doesn't make mistakes in fact it needs the mistakes to learn and it describes the process awesome points it's that operational insight that if you're getting 100 on the test like something is at the very best less than informative at the very worst you're you're way down the wrong path and um and then it reminded me of like a grocery store is it that we're ranking the different objects according to one measure no not really it's like balancing strategies for finding preference and that's where all of these assumptions like the factorization and the mean field and it's not guaranteed to give you the best object every time and there's still the stochasticity of the real world and the ability of the grocery store to change like it's not the end of the story it's actually just a practice in a process that has some of those features of reactive graphs and biological systems that's right yeah well um if any of you have any final comments this was just really one of the coolest times to learn about this you're always welcome back to join any time as a participant or to present and if you have any final comments you can note them well it was a real pleasure to be here and uh if we have new stuff we would love to come back i think it's a fantastic fantastic show here or show it's a forum or whatever it's properly called but i really enjoy it and we really enjoyed being part of it and uh thanks a lot for the discussion some some food for thought there as well it's really cool good question for every opportunity to present yes great okay until next time bye